MIL-E-51001A(MU)
11 AUGUST 1965
SUPERSEDING
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8 JUNE 1960

MILITARY SPECIFICATION

EGGS, CHICKEN, EMBRYONATED

1. SCOPE

1.1 Scope. This specification covers embryonated chicken eggs for biological use.

1.2 Classification. Embryonated chicken eggs covered by this specification shall be classified by condition as specified below.

Condition 1 — 4 days incubation.

Condition 2 — 5 days incubation.

Condition 3 — 6 days incubation.

Condition 4 — 7 days incubation.

Condition 5 — 8 days incubation.

Condition 6 — 9 days incubation.

Condition 7 — 10 days incubation.

Condition 8 — 11 days incubation.

Condition 9 — 12 days incubation.

Condition 10 — 13 days incubation.

Condition 11 — 14 days incubation.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-51049 — Eggs, Chicken, Fertile.

STANDARD

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 — Marking for Shipment and Storage.

(Copies of specifications, standards, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules, and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

FSC 1380

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UNITED STATES DEPARTMENT OF AGRICULTURE

Miscellaneous Publication No. 739 — The National Poultry and Turkey Improvement Plans and Auxiliary Provisions.

(United States Department of Agriculture publications are available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402).

3. REQUIREMENTS

3.1 Facilities and equipment.

3.1.1 *Hatchery requirements.* The hatchery shall handle only eggs from flocks that have an official state rating equivalent to U. S. Pullorum-Typhoid Clean in accordance with U. S. Department of Agriculture Miscellaneous Publication 739.

3.1.2 *Egg handling and incubation rooms.* Rooms where eggs are handled and incubated shall be clean, well ventilated, free from dust, and used solely for such purposes. Cleaning operations shall include regular use of suitable disinfectants to minimize chance of egg infection.

3.1.3 *Incubators.* Incubators used for embryonated egg supply shall not be used for hatching. They shall be kept clean at all times, and shall be disassembled and disinfected or fumigated not less than once each month or prior to embryonating eggs. (See 6.3.1.)

3.1.4 *Trays.* Trays used for incubating eggs shall be cleaned and disinfected prior to each use.

3.2 Fertile non-incubated eggs.

3.2.1 *General.* The eggs furnished for incubation shall meet the requirements of MIL–E–51049.

3.2.2 *Pre-incubation storage environmental conditions.* Eggs supplied for incubation shall have been stored air cell up, in a clean, cool egg room at a temperature of 60° ± 5°F and 70 ± 10 percent relative humidity from time of delivery to incubation.

3.2.3 *Age.* Fertile eggs used for embryo development shall be incubated within seven days after being laid.

3.3 Embryonated eggs.

3.3.1 *Environmental requirements.*

3.3.1.1 *Incubation.* The eggs shall be incubated at 99.5° ± 0.5°F and relative humidity not less than 60 per cent.

3.3.1.2 *Handling.* The eggs shall be incubated with the air cell up and shall be turned every three to four hours. No live, embryonated eggs shall be removed from the incubator until time of removal for delivery. Egg environmental temperature, from time of removal from the incubator until delivery shall be maintained at 95° ± 5°F.

3.3.2 *Viability.* All acceptable embryonated eggs shall contain living embryos when tested as specified in 4.6.1.

3.3.3 *Condition.* Egg condition (development) at time of delivery shall be within ± 12 hours of that specified in the contract or purchase order (see 6.1).

3.3.4 *Candling.* The eggs shall be candled just prior to delivery. There shall be no infertile eggs, eggs containing dead embryos, detectable contamination, or otherwise abnormal eggs, in eggs delivered, when tested as specified in 4.6.1.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

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4.1.1 *Supplier responsibility.* Unless otherwise specified in the contract or purchase order, the supplier shall be responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of examinations and tests shall be kept complete for one year, and be available to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 *Government test responsibility.* The Government will be responsible for the performance of tests specified in 4.6.1.

4.1.3 *Government inspection of facilities.* The contractor's plant and facilities shall be subject to periodic inspection by the contracting officer. The contractor shall be responsible for making the necessary arrangements so that the plant and facilities of all suppliers of eggs to the contractor may be periodically inspected by the contracting officer.

4.2 Special provisions.

4.2.1 *Alternative inspection.* The supplier may utilize any alternative inspection procedure which will provide equal or better assurance of quality by submitting a written proposal, with justification, and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification shall govern.

4.2.2 *Objective evidence.* The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, 3.3, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.3 Lotting. Original lot identification as specified in MIL–E–51049 shall be maintained throughout the specified incubation period and preparation for delivery.

4.4 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL–STD–105.

4.5 Inspection provisions.

4.5.1 *For examination and nondestructive tests.* Sample embryonated eggs and the packaging, packing, and marking requirements of section 5 shall be examined and tested in accordance with the classification of defects and MIL–STD–105.

4.5.2 *Defects.* Each embryonated egg shall be inspected for characteristics listed in the classification of defects.

4.5.3 *Classification of defects.*

4.5.3.1 *Eggs, embryonated.*

Categories	*Defects*	*Acceptance Standards*
Major:	100 percent inspection	
101	Cracked shell	
102	Egg position during incubation	
103	Handling	3.3.1.2
104	Viability	3.3.2
105	Condition (see 1.2)	3.3.3
106	Contamination	
107	Abnormality	
108	Lot identification during incubation	4.3

4.5.3.2 *Packaging, packing, and marking.*

Categories	*Defects*	*Acceptance Standards*
Major:	AQL 4.0 percent defective	
101	Improper cushioning	
102	Cases not properly marked	
103	Dirty or worn fillers or flats	

4.6 Tests.

4.6.1 *Embryo development test.* The eggs shall be candled to insure compliance with

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requirements of 3.3.2 and 3.3.5. Living embryos are motile, of a reddish color, and have visible blood vessels. Dead embryos are non-motile, yellow in color and have collapsed blood vessels. Samples from questionable lots may be selected and incubated to hatching at the discretion of the contracting officer.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Eggs shall be placed in new, standard 30-dozen cases, except when quantities delivered to a single location involve 180 eggs or less, in which event, shipment may be made in 15-dozen cases (see 6.2).

Flats used to package the eggs shall be of one piece construction and have five rows of six eggs each. No dirty or worn fillers or flats shall be used. Packaging materials shall not give off any foreign odors.

Eggs shall be placed in the flats larger end up. Any eggs not so packed shall be rejected.

5.2 Packing.

5.2.1 *Temperature in transit.* During shipment, the eggs shall be maintained at conditions specified in 3.3.1.2. Instruments shall be used to provide a record of temperature maintained in transit.

5.2.1.1 *Temperature records.* The contractor shall supply dated and certified temperature records, or copies thereof, to the Government as directed by the contracting officer.

5.2.2 *Cushioning media.* Adequate cushioning media shall be provided between the egg cases and transport vehicle bed.

5.2.3 *Case standards.* Cases, flats, fillers, filler-flats, and closure for shipment by common carrier shall meet requirements of Uniform Freight Classification, Ratings, Rules, and Regulations.

5.3 Marking. Each egg case shall be labeled with the following information.

Date of Delivery

Name of Contractor

Contract Number Condition (See 1.2)

Number of Eggs

Destination

Name or Identification No. of Source Flock

Lot Number

Each egg case shall be stamped on the exterior surface of each side with the letters "EGGS". Lettering shall be not less than three inches high.

The lid of each case shall be printed or stenciled in letters no less than one inch high with the words "THIS SIDE UP, DO NOT DROP."

In addition to the special markings specified above, all shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Condition required. (See 1.2)

(c) FOB point to be destination.

(d) Conditions of contractual eligibility.

(e) Destination test laboratory, if applicable.

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6.2 **Cases, flats, and fillers.** Cases, flats, and fillers shall conform to requirements of "KYS Filler-Flats" manufactured by Key Fiber Sales Corp., New York, New York, or equal.

6.3 **Incubator fumigation and neutralization.**

6.3.1 *Fumigation.* Incubators may be fumigated by bringing the incubator up to incubation temperature and humidity and then placing a container containing potassium permanganate and formalin in the incubator. The proportions used per 100 cubic feet of air space to be fumigated shall be 40 grams (1 1/3 ounces) of potassium permanganate to 80 cubic centimeters (2 2/3 fluid ounces) of formalin. The potassium permanganate should be added to the formalin. The mixing container should have a volume at least 10 times that of the combined ingredients. The incubator should be fumigated for no less than 20 minutes before exhausting or neutralizing the gas.

6.3.2 *Neutralization.* The gas may be neutralized after fumigation by a 33 percent ammonium hydroxide solution. The volume of ammonium hydroxide solution used should be one-half of that of the formalin used for fumigation. Neutralization may be accomplished by suspending a piece of cheese cloth saturated in the solution in the incubator, or by sprinkling the solution on the floor of the incubator.

Custodian:

Army—MU

Preparing activity:

Army—MU

Project No. 1380–A017

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